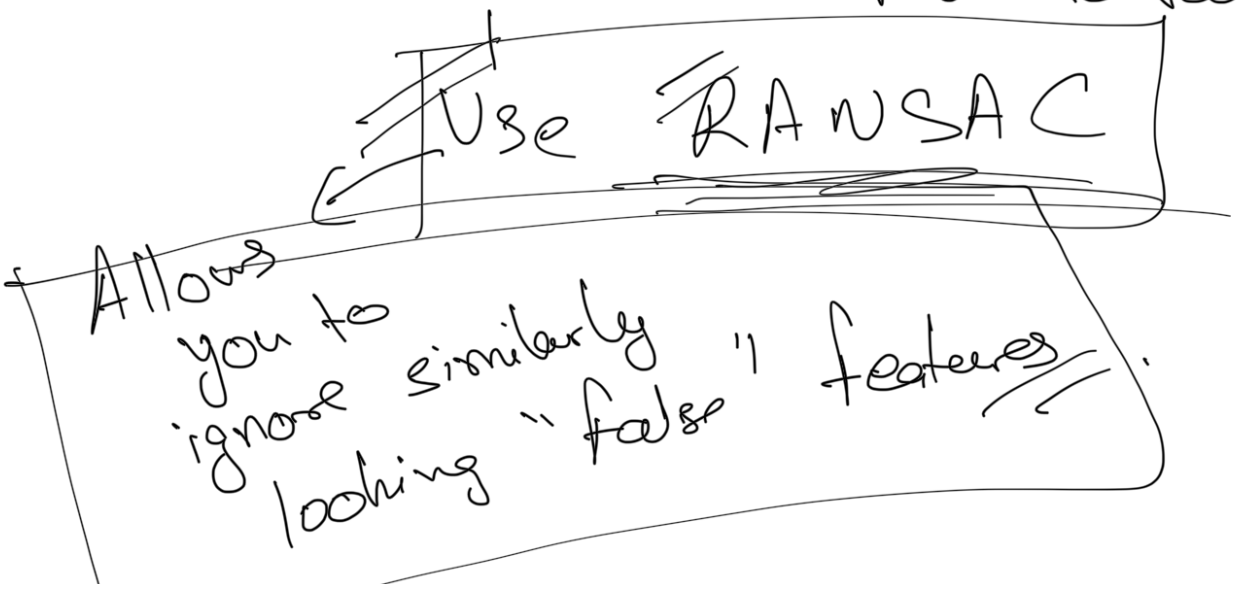
Steps for Feature Matching

- Find features (blobs, Corners, edges)
- Find descriptors
- Compare descriptors in template with descriptors in image
- Point of matching is the location of the template in image
- You have a large number of descriptors to be Compared. which descriptor to take



Flow Estimation

Object moving in scene

Image Captured

"Identify motion of the object (or estimate)" from the Image.

Applications

1) Speed Detection

2) Optical Mouse

Laser Camera (64×64) & take two images & find the shift in position

3) Camera Shake Calibration → Camera Stabilization

4) Activity Detection

5) "Gait" Recognitions



"Way a person walks"

6) 'Slow mo' videos

Motion Field

=> An object moving in 3D scene. Its image is Captured and recorded on a 2D plane.

=> The exact mapping on 2D plane is known as Motion field.



=> Motion field for a pixel => (u, v)

=> Motion field depends on the scene, Camera characteristics (intrinsic and extrinsic) And the scene being Captured.

Optical flow

This is measurable

"This is the change in intensity variation from one Snapshot to another"

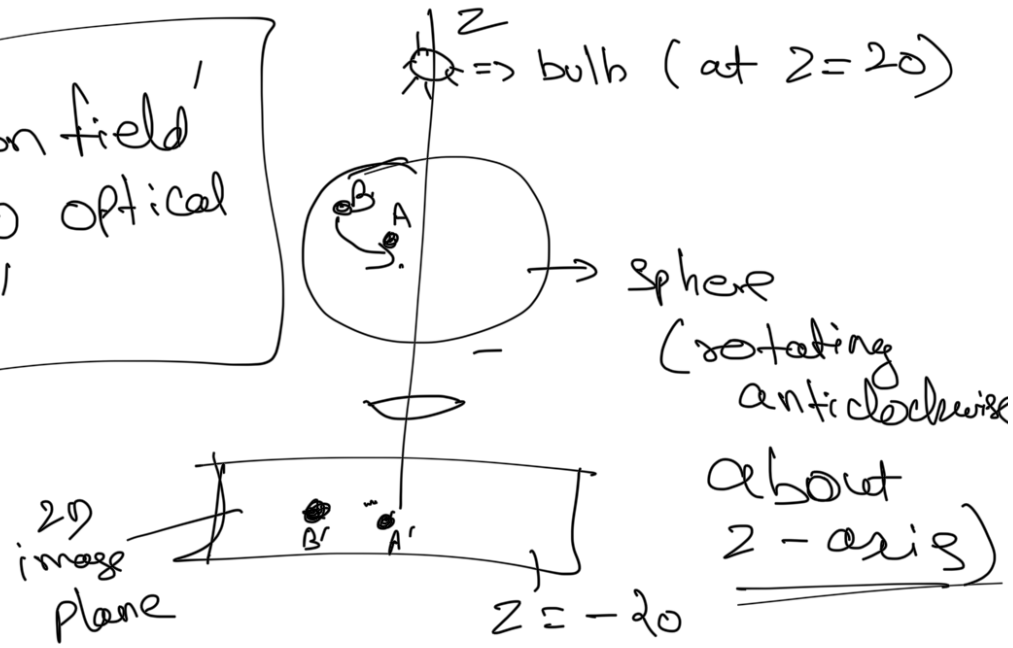
Both humans and Computers measure Optical flow.

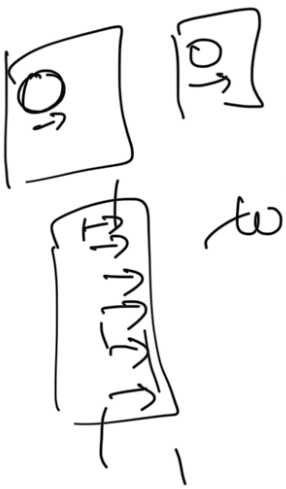
=> Optical flow gives a very good idea of Motion field in many cases.

⇒ This cannot be always estimated.

Motion field ≠ Optical flow

Has Motion field but NO optical field





what we measure always is "Optical flow"

(tracing variations in intensities or How intensities are varying).

And we hope that Optical flow will give us a good idea about motion field.

Image sitting in Cave.



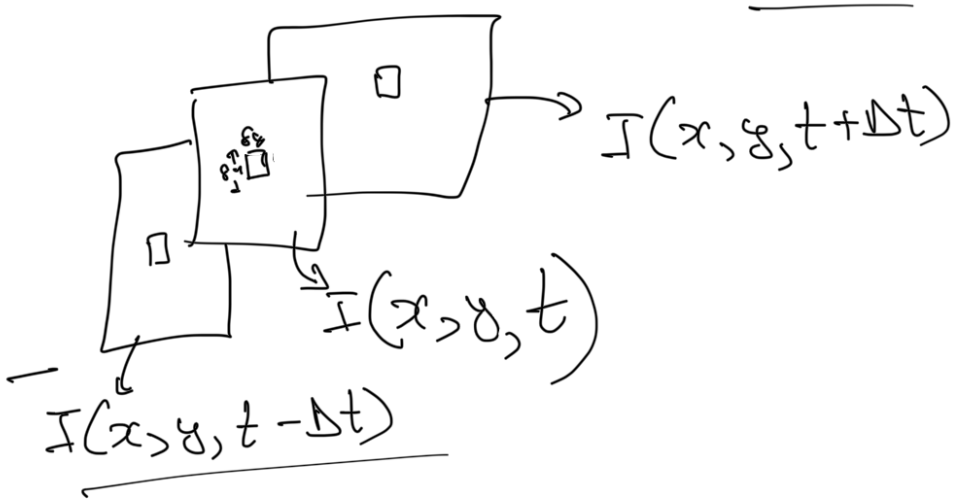
seeing trees passing by

[Q] This holds in many Practical applications

If this does not hold then some other method needs to be found out.

Optical flow Estimation

Capture snapshots of a scenery



Assumptions: For a small Δt and a small neighbourhood (x, y) to $(x+\delta x, y+\delta y)$ intensities are approximately same or constant.

No very large changes in small δt interval.

$$I(x+\delta x, y+\delta y, t+\Delta t) = I(x, y, t)$$

Assumption of brightness Constancy.

Express $I(x+\delta x, y+\delta y, t+\Delta t)$ in form of first order taylor series expansion.

$$I(x+\delta x, y+\delta y, t+\Delta t) = I(x,y,t) + \frac{\partial I}{\partial x}dx + \frac{\partial I}{\partial y}dy + \frac{\partial I}{\partial t}dt + O(dx) + O(dy) + O(dt)$$

Use Assumption of brightness Constancy

$$\frac{\partial I}{\partial x}dx + \frac{\partial I}{\partial y}dy + \frac{\partial I}{\partial t}dt = 0$$

$$\frac{\partial I}{\partial x} = I_x \quad , \quad \frac{\partial I}{\partial y} = I_y$$

$$\frac{\partial I}{\partial t} = I_t$$

Divide by dt both sides

$$I_x \frac{dx}{dt} + I_y \frac{dy}{dt} + I_t = 0$$

$$I_x u + I_y v + I_t = 0$$

Optical flow equation

Under Constrained Problem

of variables = 2

of equations = 1